Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Final Examination 2022
Course Title: Molecular Symmetry and Group Theory
Course No.: Chem-324H
Time: 2.5 hours; Full Marks: 35

[*Answer any FIVE (05) questions*]

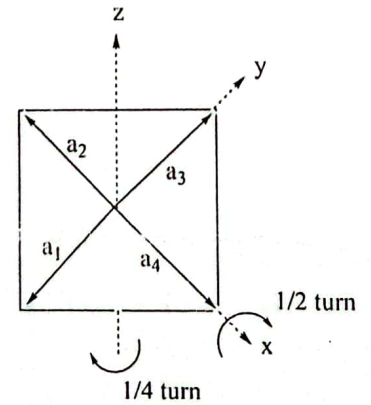
1. (a) Define '*symmetry operation*' and '*symmetry element*'? When does a molecule possess '*center of symmetry*'? **1.5 + 1 + 1.5 = 4**
(b) Show all the C_2 axes in XeF_4. **1**
(c) List the four symmetry elements of *trans*-1,2-dichloroethene. **1**
(d) Suppose a molecule belongs to the point group C_{2h}. What information about the molecular symmetry are embedded in the point group symbol C_{2h}? **1**

2. (a) Determine the point group of the following molecules using an appropriate flow chart: **1.5 × 3 = 4.5**
(i) $POCl_3$, (ii) C_6F_6, (iii) Boric acid (H_3BO_3)
(b) Set up the multiplication table for the symmetry operations of water molecule (H_2O). Apply the top operation then the side operation during the process. **2.5**

3. (a) What is the condition that needs to be fulfilled if the two symmetry operations belong to the same class. **1**
(b) Taking the C_2 axis as the z axis to what representations do the p_x, p_y and p_z orbitals belong in C_{2v} symmetry. The Character Table of C_{2v} point group is given below, in part, for your convenience: **3**

C_{2v}	E	C_2	$\sigma(xz)$	$\sigma(yz)$
A_1	1	1	1	1
A_2	1	1	-1	-1
B_1	1	-1	1	-1
B_2	1	-1	-1	1

(c) Use the following diagram to answer the questions listed below (i and ii): **2 + 1 = 3**



(i) Generate a matrix representation for quarter turn (about z axis) and half turn (about x axis) rotations using the four arrows (a_1, a_2, a_3 and a_4) shown in the above diagram as the basis.
(ii) Generate a matrix representation for the reflection in the xy plane using the same basis.

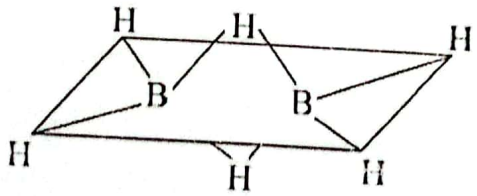
4. Find out a reducible representation taking the five P–F bonds in PF_5 molecule as the basis. Reduce it and find out the orbitals of phosphorus (P) atom that will be suitable for forming the five P–F bonds. The Character Table of D_{3h} point group is given below for your convenience: 7

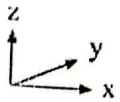
D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$		
A_1'	1	1	1	1	1	1		x^2+y^2, z^2
A_2'	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	(x^2-y^2, xy)
A_1''	1	1	1	-1	-1	-1		
A_2''	1	1	-1	-1	-1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

5. Set up the molecular orbital correlation diagram for the NH_3 molecule, considering the $2s$ and $2p$ orbitals of nitrogen and the $1s$ orbital of each hydrogen atom. The Character Table of C_{3v} point group is given below for your convenience: 7

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	-1	R_z	
E	2	-1	0	(x, y) (R_x, R_y)	(x^2-y^2, xy) (xz, yz)

6. Find the number of terminal B–H stretching vibrations which are active in the infrared and Raman spectra of diborane (B_2H_6). The Character Table of D_{2h} point group is given below for your convenience: 7

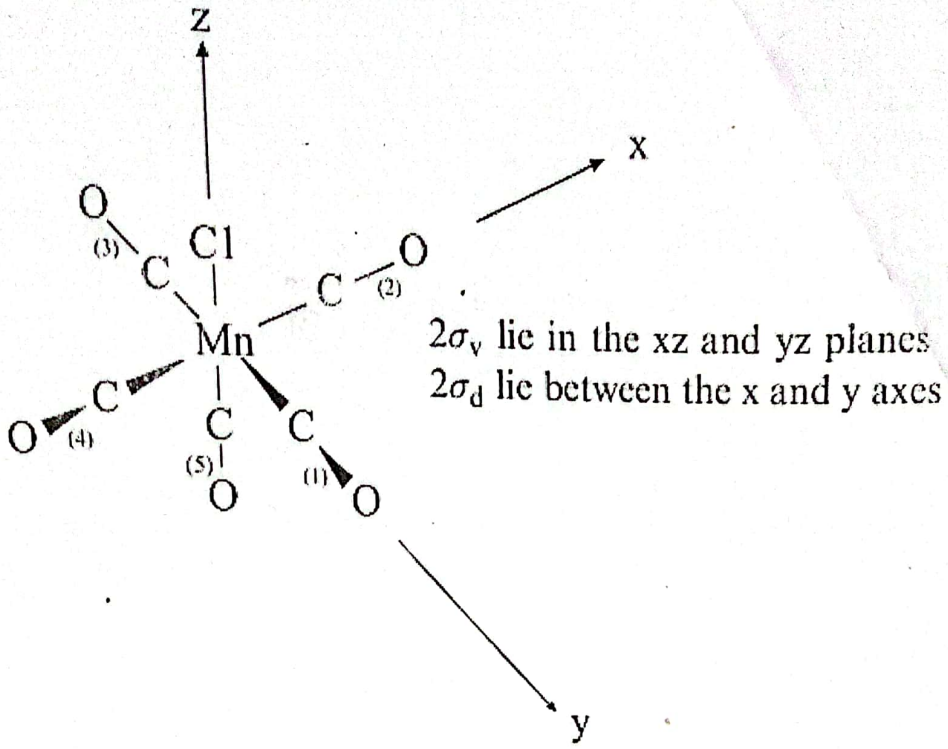




D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$		
A_g	1	1	1	1	1	1	1	1		x^2, y^2, z^2
B_{1g}	1	1	-1	-1	1	1	-1	-1	R_z	xy
B_{2g}	1	-1	1	-1	1	-1	1	-1	R_y	xz
B_{3g}	1	-1	-1	1	1	-1	-1	1	R_x	yz
A_u	1	1	1	1	-1	-1	-1	-1		
B_{1u}	1	1	-1	-1	-1	-1	1	1	z	
B_{2u}	1	-1	1	-1	-1	1	-1	1	y	
B_{3u}	1	-1	-1	1	-1	1	1	-1	x	

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7. $[Mn(CO)_5Cl]$, which belongs to C_{4v} point group, has $2A_1 + B_1 + E$ carbonyl (C–O) stretching modes.



(a) Find the normalized symmetry adapted linear combinations of the C–O stretches which have these symmetry properties. **3**

(b) Find the linear combinations of the two A_1 modes. **1**

(c) Show that your results are mutually orthogonal. **3**

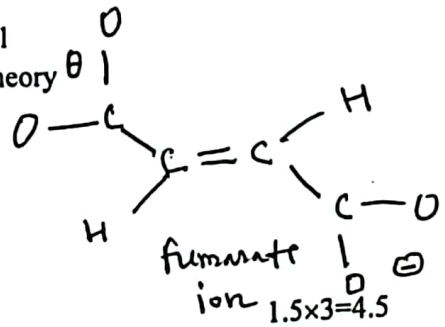
The Character Table of C_{4v} point group is given below for your convenience:

C_{4v}	E	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$		
A_1	1	1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	1	-1	-1	R_z	
B_1	1	-1	1	1	-1		x^2-y^2
B_2	1	-1	1	-1	1		xy
E	2	0	-2	0	0	(x, y) (R_x, R_y)	(xz, yz)

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Final Examination 2021
Course Title: Molecular Symmetry and Group Theory
Course # Chem. 324H
Time: 2.5 hours, Full Marks: 35



[Answer any *FIVE (05)* questions]

1. (a) Define the following terms:
(i) *Symmetry operation*; (ii) *n-fold axis of symmetry*; (iii) *Center of symmetry* 1.5×3=4.5
(b) Show the C_4 axis, σ_h and a σ_v in XeF_4. 1.5
(c) Suppose a molecule belongs to the point group D_{4h}. What information about the molecular symmetry are embedded in the point group symbol D_{4h}? 1

2. (a) Using an appropriate flow chart determine the point group of the following molecules: 2×3=6
(i) $CHCl_3$, (ii) XeF_4, (iii) Fumarate ion $(O_2CHC{=}CHCO_2)^{2-}$
(b) Write down the name of special point group to which they belong for each of the following molecules: CBr_4, SF_6, CO_2, HCl 0.25×4=1

3. Set up a molecular orbital diagram for PCl_5 molecule which belongs to D_{3h} point group? 7
The Character Table of D_{3h} point group is given below:

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$		
A_1'	1	1	1	1	1	1		x^2+y^2, z^2
A_2'	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	(x^2-y^2, xy)
A_1''	1	1	1	-1	-1	-1		
A_2''	1	1	-1	-1	-1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

4. (a) Write the 'Reduction Formula' that is used to reduce the reducible representations to their component irreducible representations. 1
(b) Taking the C_2 axis as the z axis, and σ_h to be the xy plane, to what representations do the d_{xy}, d_{xz}, and d_{yz} orbitals belong in C_{2h} symmetry. The Character Table of C_{2h} point group is given below, in part, for your convenience: 3

C_{2h}	E	C_2	i	σ_h
A_g	1	1	1	1
B_g	1	-i	i	-1
A_u	1	1	-1	-1
B_u	1	-1	-1	1

(c) Generate a matrix representation of the two-fold rotation for H_2O molecule, using the four arrows shown below as the basis (N.B. a_2 and a_3 are perpendicular to the plane, a_1 and a_2 are in it). Use the same four arrows as the basis to generate a matrix representation of the operation of reflection in the plane of the molecule. Also find the characters of these matrix representations. 3

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5. Set up the molecular orbital correlation diagram for the NH_3 molecule, considering the 2*s* and 2*p* orbitals of nitrogen and the 1*s* orbital of each hydrogen atom. The Character Table of C_{3v} point group is given below for your convenience.

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	-1	R_z	
E	2	-1	0	$(x, y)(R_x, R_y)$	$(x^2-y^2, xy)\ (xz, yz)$

6. XeF_4 (D_{4h}) has $A_{1g} + B_{1g} + E_u$ X–F stretching modes:
   (a) Find the normalized symmetry adapted linear combination of the Xe-F stretches which have these symmetry properties. 3
   (b) Find linear combination of the two E_u modes. 1
   (c) Show that your results are mutually orthogonal. 3

   The Character Table of D_{4h} point group is given below:

D_{4h}	E	$2C_4$	C_2	$2C_2'$	$2C_2''$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	1	-1	-1	1	1	1	-1	-1	R_z	
B_{1g}	1	-1	1	1	-1	1	-1	1	1	-1		x^2-y^2
B_{2g}	1	-1	1	-1	1	1	-1	1	-1	1		xy
E_g	2	0	-2	0	0	2	0	-2	0	0	(R_x, R_y)	(xz, yz)
A_{1u}	1	1	1	1	1	-1	-1	-1	-1	-1		
A_{2u}	1	1	1	-1	-1	-1	-1	-1	1	1	z	
B_{1u}	1	-1	1	1	-1	-1	1	-1	-1	1		
B_{2u}	1	-1	1	-1	1	-1	1	-1	1	-1		
E_u	2	0	-2	0	0	-2	0	2	0	0	(x, y)	

7. Find the number and symmetry species of the Raman and infrared active vibrations of boron trichloride (D_{3h}). The Character Table of D_{3h} point group is given below: 7

BCl_3

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$		
A_1'	1	1	1	1	1	1		x^2+y^2, z^2
A_2'	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	(x^2-y^2, xy)
A_1''	1	1	1	-1	-1	-1		
A_2''	1	1	-1	-1	-1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

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$2E' + A_2''$

$-1 + 2$

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2020
Course Title: Molecular Symmetry and Group Theory
Course # Chem. 324, Full Marks: 35, Time: 2.5 hrs.

[Answer any five questions]

1. (a) Define the following terms: 1.5×3
(i) *Symmetry element*; (ii) *n-fold improper axis of symmetry* and
(iii) *Plane of symmetry*
(b) Show the C_3 axis, σ_h and a σ_v in PF_5. 1.5
(c) What is the difference between vertical plane and horizontal plane? 1.0

2. (a) Using an appropriate flow chart determine the point group of the following molecules: 2×3
(i) $POCl_3$, (ii) BF_3 and (iii) $Ni(CO)_4$
(b) Suppose a molecule belongs to the point group C_{2v}. What are the meanings of the point group symbol C_{2v}? 1.0

3. (a) What rules should be followed by a set of symmetry operations of any geometrical shape to form a mathematical group? 2.0
(b) Taking the C_2 axis as the z axis, and σ_h to be the xy plane, to what representations do the P_x, P_y, and P_z orbitals belong in C_{2h} symmetry (the Character Table of C_{2h} point group is given below: 3.0

C_{2h}	E	C_2	i	σh
A_g	1	1	1	1
B_g	1	-1	1	-1
A_u	1	1	-1	-1
B_u	1	-1	-1	1

(c) Set up the matrices which will perform the following transformations: 2.0

(A) $\begin{pmatrix} x \\ y \end{pmatrix}$ to $\begin{pmatrix} -y \\ -x \end{pmatrix}$ (B) $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ to $\begin{pmatrix} -y \\ x \\ -z \end{pmatrix}$

4. Find the hybrid orbitals of boron atom suitable for forming the three B–H bonds in BH_3 molecule (the Character Table of D_{3h} point group is given below: 7.0

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$		
A_1'	1	1	1	1	1	1		x^2+y^2, z^2
A_2'	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	(x^2-y^2, xy)
A_1''	1	1	1	-1	-1	-1		
A_2''	1	1	-1	-1	-1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

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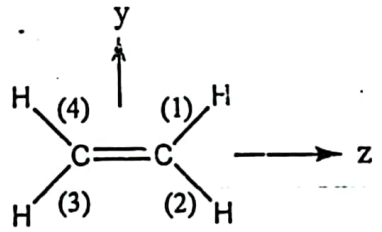
5. Set up the molecular orbital correlation diagram for the CH_4 molecule, considering the 2*s* and 2*p* orbitals of carbon and the 1*s* orbital of each hydrogen atom (the Character Table of T*d* point group is given below): 7.0

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$		
A_1	1	1	1	1	1		$x^2+y^2+z^2$
A_2	1	1	1	-1	-1		
E	2	-1	2	0	0		$(2z^2-x^2-y^2, x^2-y^2)$
T_1	3	0	-1	1	-1	(R_x, R_y, R_z)	
T_2	3	0	-1	-1	1	(x, y, z)	

6. Find the number and symmetry species of the Raman and infrared active C≡O stretching vibrations of *fac*-$[CrL_3(CO)_3]$ (the Character Table of C_{3v} point group is given below: 7.0

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	-1	R_z	
E	2	-1	0	(x, y) (R_x, R_y)	(x^2-y^2, xy) (xz, yz)

7. (a) Find the symmetry adapted linear combinations of the C–H bond stretches of ethane (D_{2h}) and hence find the modes of vibration of the C–H bonds. Use the following co-ordinate system (the Character Table of D_{2h} point group is given below): 5.0



(b) Normalize your results. 2.0

D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$		
A_g	1	1	1	1	1	1	1	1		x^2, y, z^2
B_{1g}	1	1	-1	-1	1	1	-1	-1	R_z	xy
B_{2g}	1	-1	1	-1	1	-1	1	-1	R_y	xz
B_{3g}	1	-1	-1	1	1	-1	-1	1	R_x	yz
A_u	1	1	-1	1	-1	-1	-1	-1		
B_{1u}	1	1	-1	-1	-1	-1	1	1	z	
B_{2u}	1	-1	1	-1	-1	1	-1	1	y	
B_{3u}	1	-1	-1	1	-1	1	1	-1	x	

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Department of Chemistry, Jahangirnagar University
Course Title: Molecular Symmetry and Group Theory
Course no.: Chem. 324; Marks: 10, Time: 35 min.

(a) Define the following terms: $1.25 \times 2 = 2.5$

(i) *Proper rotation axis*; (ii) *Improper rotation axis*

(b) Using a flow chart determine the point group of boric acid (H_3BO_3), chloroform ($CHCl_3$) and benzene (C_6H_6). $1.5 \times 3 = 4.5$

(c) What is the condition that needs to be fullfilled if the two symmetry operations belong to the same class. 1

(d) Taking the C_2 axis as the z axis, and σ_h to be the xy plane, to what representations do the x, y, and z belong in C_{2h} symmetry (the Character Table of C_{2h} point group is given below, in part, for your convenience). 2

C_{2h}	E	C_2	i	σ_h	
A_g	1	1	1	1	
B_g	1	-1	1	-1	
A_u	1	1	-1	-1	z
B_u	1	-1	-1	1	x, y

Department of Chemistry, Jahangirnagar University
Course Title: Molecular Symmetry and Group Theory
Course no.: Chem. 324; Marks: 10, Time: 45 min.

1. Set up the molecular orbital correlation diagram for the PF_5 molecule, considering the 3*s*, 3*p* and 3*d* orbitals of phosphorus and the singly occupied *p* orbital of each fluorine atom. The Character Table of D_{3h} point group is given below for your convenience:

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$		
A_1'	1	1	1	1	1	1		x^2+y^2, z^2
A_2'	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	(x^2-y^2, xy)
A_1''	1	-1	1	-1	-1	-1		
A_2''	1	1	-1	-1	1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

1. Find the number of C–O stretching vibrations which are active in the infrared and Raman spectra of $[M(CO)_3L_3]$. The Character Table of C_{3v} point group is given below for your convenience: 4

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	-1	R_z	
E	2	-1	0	(x, y) (R_x, R_y)	(x^2-y^2, xy) (xz, yz)

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Department of Chemistry, Jahangirnagar University
Course Title: Molecular Symmetry and Group Theory
Course no.: Chem. 324; Marks: 10, Time: 1 h.

1. Set up the molecular orbital correlation diagram for the CH_4 molecule, considering the 2*s* and 2*p* orbitals of carbon and the 1*s* orbital of each hydrogen atom (the Character Table of T_d point group is given below for your convenience). 5

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$		
A_1	1	1	1	1	1		$x^2+y^2+z^2$
A_2	1	1	1	-1	-1		$(2z^2-x^2-y^2, x^2-y^2)$
E	2	-1	2	0	0		
T_1	3	0	-1	1	-1	(R_x, R_y, R_z)	
T_2	3	0	-1	-1	1	(x, y, z)	

2. Find the number and symmetry species of the Raman and infrared active vibrations of CO_3^{2-} ion. (the Character Table of D_{3h} point group is given below for your convenience). 5

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$		
A_1'	1	1	1	1	1	1		x^2+y^2, z^2
A_2'	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	(x^2-y^2, xy)
A_1''	1	1	1	-1	-1	-1		
A_2''	1	1	-1	-1	-1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

Department of Chemistry, Jahangirnagar University
Course Title: Molecular Symmetry and Group Theory
Course no.: Chem. 324; Marks: 10, Time: 45 min.

1. (a) Define the following terms: $1.5 \times 3 = 4.5$
(i) *Symmetry Operation*; (ii) *Plane of symmetry*; (iii) *Center of Symmetry*
(b) Draw the structure of XeF_4 and show the C_4 axis, σ_h and a σ_v in it. 1.5
(c) Using a flow chart determine the point group of $POCl_3$ and PF_5. $2 \times 2 = 4$

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Department of Chemistry, Jahangirnagar University
Course Title: Molecular Symmetry and Group Theory
Course no.: Chem. 324; Marks: 10, Time: 35 min.

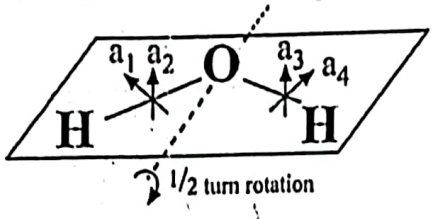
1. (a) Define the following terms: $1.5 \times 2 = 3$
(i) *Symmetry Operation*; (ii) *Center of Symmetry*
(b) Using a flow chart determine the point group of B_2H_6 and PCl_5. $2 \times 2 = 4$
(c) Taking the C_2 axis as the z axis, and σ_h to be the xy plane, to what representations do the P_x, P_y, and P_z orbitals belong in C_{2h} symmetry (the Character Table of C_{2h} point group is given below, in part, for your convenience). 3

C_{2h}	E	C_2	i	σ_h
A_g	1	1	1	1
B_g	1	-1	1	-1
A_u	1	1	-1	-1
B_u	1	-1	-1	1

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Department of Chemistry, Jahangirnagar University
Course Title: Molecular Symmetry and Group Theory
Course no.: Chem. 324; Marks: 10, Time: 45 min.

1. Generate a matrix representation of the two fold rotation for H_2O molecule, using the four arrows shown as the basis (N. B. a_2 and a_3 are perpendicular to the plane, a_1 and a_2 are in it). Use the same four arrows as the basis to generate a matrix representation of the operation of reflection in the plane of the molecule. Also find the characters of these matrix representations. 3



2. Set up the molecular orbital correlation diagram for the CH_4 molecule, considering the $2s$ and $2p$ orbitals of carbon and the $1s$ orbital of each hydrogen atom. Also find the hybrid orbitals of carbon atom suitable for forming the four C–H bonds in CH_4 molecule (the Character Table of T_d point group is given below for your convenience). 7

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$		
A_1	1	1	1	1	1		$x^2+y^2+z^2$
A_2	1	1	1	-1	-1		
E	2	-1	2	0	0		$(2z^2-x^2-y^2, x^2-y^2)$
T_1	3	0	-1	1	-1	(R_x, R_y, R_z)	
T_2	3	0	-1	-1	1	(x, y, z)	

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